RecoMart Recommendation System: Problem Formulation & Architecture

Prepared by RecoMart Data Platform Team | Data Engineering & ML Architecture

1. Business Objective & Context

RecoMart is a rapidly growing e-commerce platform aiming to maximize user engagement, click-through rates (CTR), and average order value (AOV) via hyper-personalized product recommendations. By deploying a robust end-to-end data and ML pipeline, RecoMart can systematically ingest multi-source user interactions, curate validated features, and continuously update recommendation models.

2. Data Sources & Attributes

Data Source	Format / Ingestion Mode	Key Attributes Description
Clickstream & Transactions	CSV (Batch Ingestion)	user_id, item_id, rating (1.0-5.0), interaction_type (view, cart, purchase), timestamp, device
Product Metadata	REST API (Batch/Near-RT)	item_id, product_name, category, brand, price, sentiment_score, popularity_score
User Demographics	CSV / DB	user_id, age, gender, signup_date, membership_tier

3. Target Pipeline Outputs

• **Raw Data Lake:** Partitioned storage by data source, year, month, and day.

• **Validated Datasets:** Automated quality reports flagging nulls, schema drifts, and out-of-bound ratings.

• **Feature Store:** Standardized user and item offline/online feature views for zero-leakage training.

• **Trained Recommendation Models:** Collaborative filtering (SVD) and content-based recommendation artifacts tracked via MLflow.

• **Inference API:** High-throughput scoring interface for real-time recommendation retrieval.

4. Evaluation Metrics

Metric	Category	Target Benchmark / Description
Precision@K	Ranking Quality	Fraction of recommended items in Top-K that are relevant to the user.
Recall@K	Coverage	Fraction of total relevant items retrieved within Top-K.
NDCG@K	Ranking Order	Normalized Discounted Cumulative Gain accounting for position decay.
RMSE / MAE	Rating Error	Root Mean Squared Error & Mean Absolute Error on predicted rating scores.